

A POLYNOMIAL SCALING WINDOW FOR RANDOM k -SAT AND A PROOF OF THE SATISFIABILITY CONJECTURE

GAIA CARENINI

ABSTRACT. We prove that the scaling window of random k -SAT has width $O(n^{1/2+1/k})$, a polynomial improvement over our previous bound of $O(n/\log n)$. Combined with a result of Abbe and Montanari, this establishes the satisfiability conjecture for every fixed $k \geq 3$: there exists a constant α_k such that a random k -SAT formula on n variables with $\lfloor \alpha n \rfloor$ clauses is satisfiable with high probability if $\alpha < \alpha_k$ and unsatisfiable with high probability if $\alpha > \alpha_k$.

1. INTRODUCTION

How many random clauses are needed to make a Boolean formula unsatisfiable? For random k -SAT, a sharp transition is known to occur, but sharpness and convergence of the transition density are different questions. The purpose of this paper is to obtain a sufficiently strong quantitative form of sharpness to settle the latter question as well.

Let $\mathcal{C}_n$ be the set of signed k -clauses on $[n] = \{1, \dots, n\}$, each with k distinct variables, and let $N_n = |\mathcal{C}_n| = 2^k \binom{n}{k}$. This is the Boolean case of the notation in [7], with the fixed alphabet size and clause length suppressed. Let $F_{n,m}$ be the conjunction of m independent uniform members of $\mathcal{C}_n$; repetitions between clauses are allowed. The parameter $\alpha = m/n$ is the *clause density*. Throughout the paper, k is fixed and $n \rightarrow \infty$, and an event holds *with high probability* if its probability tends to one.

Early experiments revealed a pronounced transition in the satisfiability of random formulas as the clause density increases. Mitchell, Selman and Levesque [12] observed that the computational effort of the satisfiability procedure they tested peaked near this transition. These observations helped motivate two closely related lines of research: determining when a random formula has a solution, and understanding the difficulty of finding or certifying the absence of one.

The *satisfiability conjecture* asserts that, for every fixed $k \geq 3$, there is a constant α_k such that $F_{n, \lfloor \alpha n \rfloor}$ is satisfiable with high probability for $\alpha < \alpha_k$ and unsatisfiable with high probability for $\alpha > \alpha_k$. Thus the transition is conjectured to occur at a limiting clause density independent of n . In this paper, we prove the conjecture.

A major advance towards the conjecture was Friedgut's sharp-threshold theorem [10], with an appendix by Bourgain. For every fixed $k \geq 3$, it gives a sequence $\alpha_k(n)$, bounded above and bounded away from zero, such that, for every fixed $\eta > 0$, formulas of density $\alpha_k(n) - \eta$ are satisfiable with high probability, whereas formulas of density $\alpha_k(n) + \eta$ are unsatisfiable with high probability. The remaining issue is convergence of $\alpha_k(n)$. Sharpness alone does not exclude the possibility that the location of the transition varies with n without approaching a limit.

For 2-SAT, the limiting threshold is $\alpha_2 = 1$, and Bollobás, Borgs, Chayes, Kim and Wilson [6] determined the scaling window to have width $\Theta(n^{2/3})$ clauses. For larger k , first- and second-moment arguments gave increasingly precise bounds on the critical density [2, 3, 4]. In particular, Achlioptas and Peres [4] located the transition within an additive $O(k)$ of $2^k \log 2$. Such estimates describe the threshold as k grows, while leaving open its convergence as $n \rightarrow \infty$ for each fixed k .

Ding, Sly and Sun [8] proved the satisfiability conjecture for all sufficiently large k . They also identified the limiting threshold with the value predicted by the one-step replica symmetry breaking cavity method from statistical physics. Their proof develops a detailed analysis of the organisation of solutions into clusters, combining moment calculations with conditioning on local neighbourhood statistics and the analysis of recursions on trees. These methods establish both the existence and

the value of the threshold, but involve substantial probabilistic and analytic machinery whose extension to every fixed $k \geq 3$ presents significant difficulties.

A different route is suggested by Abbe and Montanari [1], who connect the width of the transition to convergence of its location. We retain the window notation of [7]. Let $\text{Cover}_n \subseteq 2^{\mathcal{C}_n}$ be the event of unsatisfiability, and let μ_p include each clause independently with probability p . For $u \in (0, 1)$, set $p_u(n) = \inf\{p : \mu_p(\text{Cover}_n) \geq u\}$. For $\eta \in (0, 1/2)$, the independent-inclusion window, measured in expected clause count, is $W_\eta(n) = N_n(p_{1-\eta}(n) - p_\eta(n))$. For sampling with replacement, let $\tilde{m}_u(n)$ be the least m for which $\mathbb{P}(F_{n,m} \text{ is unsatisfiable}) \geq u$, and set $\tilde{W}_\eta(n) = \tilde{m}_{1-\eta}(n) - \tilde{m}_\eta(n)$.

The Abbe–Montanari criterion, in the form stated in [13, Proposition 14], implies that the satisfiability conjecture follows if, for some fixed $\delta > 0$ and every fixed $\eta \in (0, 1/2)$,

$$\tilde{W}_\eta(n) = O_{k,\eta}\left(\frac{n}{(\log n)^{1+\delta}}\right). \quad (1.1)$$

The criterion is stated for the independent-inclusion model; Section 4 gives the transfer between the two models. Thus a sufficiently strong quantitative improvement in sharpness establishes a limiting threshold density.

Obtaining such an improvement has proved difficult. The Friedgut–Bourgain argument gives $\tilde{W}_\eta(n) = O_{k,\eta}(n/\log \log n)$. In recent work [7], we improved this to $O_{k,\eta}(n/\log n)$ as an application of a general theorem for sparse monotone covering problems. That argument combines the strengthening of Bourgain’s theorem by Keevash, Lifshitz, Long and Minzer [11] with a comparison between the effects of prescribed local constraints and additional random constraints. It also gives windows of the same order for random signed NAE- k -SAT and hypergraph non-two-colourability. The bound for k -SAT nevertheless falls short of (1.1). Here we strengthen the bound for random k -SAT to a polynomial scale.

Theorem 1.1. *For every fixed $k \geq 3$ and $\eta \in (0, 1/2)$, $\tilde{W}_\eta(n) = O_{k,\eta}(n^{1/2+1/k})$.*

For example, the bound is $O_\eta(n^{5/6})$ for 3-SAT and $O_\eta(n^{3/4})$ for 4-SAT. Since $1/2 + 1/k < 1$ for every fixed $k \geq 3$, this is a polynomial improvement over $n/\log n$ and satisfies (1.1). Combining Theorem 1.1 with the Abbe–Montanari criterion yields the following consequence.

Corollary 1.2 (Satisfiability conjecture). *For every fixed $k \geq 3$, there exists $\alpha_k \in (0, \infty)$ such that, for every fixed $\alpha \geq 0$ with $\alpha \neq \alpha_k$,*

$$\lim_{n \rightarrow \infty} \mathbb{P}(F_{n, \lfloor \alpha n \rfloor} \text{ is satisfiable}) = \begin{cases} 1, & \alpha < \alpha_k, \\ 0, & \alpha > \alpha_k. \end{cases}$$

To prove the window estimate, we couple all clause counts on one probability space. Choose an infinite sequence $C_1, C_2, \dots$ of independent uniform signed k -clauses and let $F_{n,m} = C_1 \wedge \dots \wedge C_m$, with $F_{n,0}$ the empty formula. Clauses are added one at a time, and once the formula becomes unsatisfiable it remains so. Let T_n be the first unsatisfiable clause count and put $a_n = \mathbb{E}T_n$. Since $\mathbb{P}(F_{n,m} \text{ is unsatisfiable}) = \mathbb{P}(T_n \leq m)$, the window is a quantile interval for T_n . We prove the following stronger statement.

Theorem 1.3. *For every fixed $k \geq 3$, there exists $C_k < \infty$ such that $\text{Var}(T_n) \leq C_k n^{1+2/k}$. Consequently, for every $x > 0$,*

$$\mathbb{P}(|T_n - a_n| \geq x) \leq \min \left\{ 1, \frac{C_k n^{1+2/k}}{x^2} \right\}.$$

Theorem 1.1 follows immediately by Chebyshev’s inequality. The variance estimate also controls probabilities tending to zero, beyond the fixed probability levels used to define the scaling window.

Wilson [14, Corollary 4] proved the lower bound $\tilde{W}_\eta(n) = \Omega_{k,\eta}(n^{1/2})$, including in the fixed-clause model. Our upper bound leaves a gap of $1/k$ between the exponents. Section 5 discusses this gap and the two estimates from which our exponent arises.

1.1. Motivation and the change of approach. The starting point is the local comparison in our earlier work [7], rather than the sharp-threshold theorem in which it was used. In that argument, a wide transition leads, through the quantitative Bourgain theorem of Keevash, Lifshitz, Long and Minzer [11], to a small prescribed collection of constraints whose addition increases the probability of unsatisfiability. The replacement step reproduces part of this increase using genuinely random constraints. At a scale $K \asymp n/W_\eta(n)$, the guaranteed increase is exponentially small in K , and its simulation uses at most exponentially many random constraints. The general random-coordinate increment bound then gives $e^{-CK} \lesssim e^{C'K}/\sqrt{n}$. This forces $K = \Omega(\log n)$, and hence gives $O(n/\log n)$ for the window. Changing constants or repeating the same comparison does not give another power of the logarithm. This is a limitation of that quantitative argument, not a barrier to every possible use of Friedgut–Bourgain theory.

The replacement principle contains more information than this application uses. Its underlying comparison between shorter clauses and k -clauses holds for every residual set of assignments; no booster is needed to state it. Moreover, shorter clauses arise whenever we fix a variable. This suggests applying the comparison to the two possible values of a single variable, and doing so throughout the clause process rather than only at a density selected by a sharp-threshold theorem.

The natural probabilistic framework for this change of approach is that of structured coupon collectors. Each clause covers the assignments that falsify it, so unsatisfiability occurs precisely when these random codimension- k subcubes cover $\{0, 1\}^n$. This interpretation is discussed explicitly by Falgas-Ravry, Larsson and Markström [9, Section 5.4]. The same paper recalls Aldous’s concentration criterion [9, Section 2.2, Theorem 2]: a cover time is concentrated when the expected time to cover its last uncovered set using a fresh independent continuation is small compared with the expected cover time. In the clause process, this last uncovered set is exactly the solution set of the last satisfiable formula. Aldous’s quantitative inequality [5, Theorem 1(b)] therefore suggests estimating the remaining lifetime of this formula, rather than seeking a local witness through a sharp-threshold theorem.

Two choices make this useful. Forced variables describe exactly how a single clause can eliminate the remaining solutions. The expected remaining time describes how resistant those solutions are to a fresh sequence of clauses, and has a simple conditional drift. Aldous’s covering inequality connects the latter quantity to the variance of T_n . The argument below explains how deleting one variable connects the two quantities to each other.

1.2. Overview of the proof. We shall not try to describe the whole set of satisfying assignments. Instead, we ask how long that set would survive if we continued adding independent random clauses. The proof turns this remaining lifetime into an estimate for the fluctuations of T_n .

Why look at the last satisfiable formula? For a satisfiable formula G on n variables, let $r_n(G)$ be the expected number of further independent uniform k -clauses needed to make it unsatisfiable; set $r_n(G) = 0$ if it is already unsatisfiable. The elementary first-moment bound gives $r_n(G) \leq a_n = O_k(n)$. Let $\Gamma_n = F_{n, T_n-1}$. Although the next clause in the observed process destroys Γ_n , this does not show that $r_n(\Gamma_n)$ is small: the observed clause has been selected by the fact that it succeeded, whereas r_n is defined using a fresh independent continuation.

Each clause excludes a codimension- k subcube of $\{0, 1\}^n$, and unsatisfiability means that these subcubes cover the whole cube. Aldous’s inequality [5, Theorem 1(b)] gives $\text{Var}(T_n) \leq \kappa a_n \mathbb{E} r_n(\Gamma_n)$. Thus it is enough to prove

$$\mathbb{E} r_n(\Gamma_n) = O_k(n^{2/k}). \quad (1.2)$$

The question about the whole process has become a question about the difficulty of eliminating its last surviving solutions.

What does the final clause tell us? Suppose, for example, that adjoining $x_1 \vee \neg x_2 \vee x_3$ makes a satisfiable formula unsatisfiable. Every original solution must have $x_1 = 0$, $x_2 = 1$ and $x_3 = 0$. Thus every variable in a clause that ends the process was already forced. Here a variable is *forced*

if it has the same value in every satisfying assignment, not merely in the assignments belonging to one cluster.

As in the cubical notation of [7], write $\ell(G)$ for the number of coordinates fixed throughout the solution set, and $\rho_j(G)$ for the probability that one uniform signed j -clause destroys G . We apply these symbols directly to a formula rather than naming its solution set separately. Then $\rho_j(G) = \binom{\ell(G)}{j} / (2^j \binom{n}{j})$. The clause must choose only forced coordinates and then choose the unique sign pattern disagreeing with their forced values. The solution set may be complicated, but its probability of being destroyed by one clause is determined by a single integer.

We can now identify the estimate we need. Besides $r_n(G) \leq a_n$, we have $r_n(G) \leq 1/\rho_k(G)$ whenever $\rho_k(G) > 0$: simply wait for a clause that eliminates every original solution, ignoring any help from intervening clauses. Consequently,

$$\rho_k(G)r_n(G) \leq \min\{1, a_n\rho_k(G)\} \leq \frac{a_n^{1/k}}{2} \frac{\ell(G)}{n}.$$

A satisfiable state is selected as the last one with conditional probability $\rho_k(G)$. Summing over the possible times of the final transition therefore shows that (1.2) follows from

$$\sum_{m \geq 0} \mathbb{E} \left[\mathbf{1}_{\{T_n > m\}} \frac{\ell(F_{n,m})}{n} \right] = O_k(n^{1/k}). \quad (1.3)$$

This is an integrated estimate: it bounds the expected total time for which a uniformly chosen variable is forced before unsatisfiability. We do not need to show that every formula encountered by the process has few forced variables.

How can we recognise a forced variable? Fix a variable x and delete all clauses containing it, leaving a formula G on the other $n - 1$ variables. Each choice $x = 0$ or $x = 1$ restores some of the deleted clauses as $(k - 1)$ -clauses, while satisfying the others automatically. Exactly one branch survives when the original formula is satisfiable and forces x .

Let clauses arrive according to a rate-one Poisson process. At time t , the clauses avoiding x , those containing x , and those containing $\neg x$ form independent families. Conditional on G , the two branches are therefore independent and identically distributed. Each adjoins a Poisson number of shorter clauses of mean $z = kt/(2n)$. If $q_G(z)$ is its probability of failure, then the conditional probability that the full formula is satisfiable and forces x is $2q_G(z)(1 - q_G(z))$ for satisfiable G . Deletion turns a statement about all solutions into a comparison between two independent perturbations of a common formula.

For $t = O_k(n)$ the mean z is bounded. The replacement lemma gives $q_G(z) \leq C_{k,L}r_{n-1}(G)^{-(k-1)/k}$ for $0 \leq z \leq L$. The power comes from the exact formula for ρ_j : up to finite-size corrections, ρ_k is the $k/(k - 1)$ -th power of ρ_{k-1} . Several shorter clauses may nevertheless cooperate, so a one-step comparison at the initial formula is not enough. We add both clause lengths, apply the comparison at every intermediate state, and integrate while replacing shorter clauses by ordinary ones. If $r = r_{n-1}(G)$, an independent-block argument bounds the probability of failure after time t by t/r . The interpolation costs $O_{k,L}(t^{-(k-1)})$, apart from a finite-size error. Taking $t = r^{1/k}$ balances these two terms and gives the stated power of r .

Why does the contribution not accumulate too much? The advantage of r_n is the exact first-step identity $\mathbb{E}[r_n(H \wedge C)] = r_n(H) - 1$ for satisfiable H . If r_n decreased deterministically by one at each step, summing $r_n^{-(k-1)/k}$ would cost only $O_k(a_n^{1/k})$. It need not behave in this way, but the concave potential $\Phi(r) = \int_0^r (1 + u)^{-(k-1)/k} du$ gives the same bound in expectation. Each contribution is bounded by a constant multiple of the expected decrease of Φ ; the decreases telescope. No concentration of r_n itself is required.

The deleted formula has the law of the same clause process on $n - 1$ variables, with time rescaled by $1 - k/n$. We may therefore apply this occupation estimate to it. The deletion identity and replacement lemma give (1.3) up to any sufficiently large constant multiple of n , and the

first-moment bound controls the remaining tail. Poisson time preserves the discrete occupation sums exactly.

There are now two factors of $n^{1/k}$: one from total forcing and one from the conversion to the terminal distribution. They give $\mathbb{E}r_n(\Gamma_n) = O_k(n^{2/k})$, and Aldous's inequality yields the variance bound. Chebyshev gives the window. Convergence of the threshold density is a separate final step, obtained from the Abbe–Montanari criterion after transferring the estimate to the independent-inclusion model.

Section 2 proves the replacement lemma; Section 3 gives the occupation and deletion estimates; and Section 4 completes the variance bound and the passage to a limiting density. Section 5 considers the remaining gap to the lower bound.

2. REMAINING EXPECTED TIME AND REPLACEMENT OF SHORTER CLAUSES

Fix $k \geq 3$. All logarithms are natural, and constants may depend on k . We use the clause family $\mathcal{C}_n$, the process $F_{n,m}$, and the mean $a_n = \mathbb{E}T_n$ defined above. For an arbitrary satisfiable formula G on $[n]$, define

$$r_n(G) = \mathbb{E} \min\{m \geq 1 : G \wedge C_1 \wedge \cdots \wedge C_m \text{ is unsatisfiable}\},$$

where the clauses are independent of G , and set $r_n(G) = 0$ if G is unsatisfiable. The formula G need not be a k -CNF; this allows the same definition after adding shorter clauses.

Expose the clauses at the arrival times of an independent rate-one Poisson process $(\Pi_t)_{t \geq 0}$, and write $F_n(t) = F_{n, \Pi_t}$. Let τ_n be its first unsatisfiable time and $s_n(t) = \mathbb{P}(\tau_n > t)$. Conditional on T_n , the variable τ_n is a sum of T_n independent exponential random variables of mean one. Consequently,

$$\mathbb{E}\tau_n = a_n, \quad \text{Var}(\tau_n) = \text{Var}(T_n) + a_n. \quad (2.1)$$

Every fixed assignment satisfies a uniform signed k -clause with probability $1 - 2^{-k}$. A union bound over assignments therefore gives

$$\mathbb{P}(T_n > m) \leq 2^n(1 - 2^{-k})^m, \quad s_n(t) \leq \min\{1, 2^n e^{-t/2^k}\}. \quad (2.2)$$

In particular, the hitting times have finite moments. Integrating the second bound and using the inclusion of the solution set of G in $\{0, 1\}^n$ gives

$$1 \leq r_n(G) \leq a_n \leq 2^k(n \log 2 + 1) \quad (G \text{ satisfiable}). \quad (2.3)$$

Conditioning on the first additional clause yields

$$\mathbb{E}[r_n(G \wedge C)] = r_n(G) - 1 \quad (G \text{ satisfiable}), \quad (2.4)$$

where C is a uniform member of $\mathcal{C}_n$.

For a satisfiable formula H , recall that $\ell(H)$ counts forced variables and $\rho_j(H)$ is the probability that an independent uniform signed j -clause makes H unsatisfiable, for $1 \leq j \leq k$. The clause must use only forced variables and disagree with their values, so

$$\rho_j(H) = \frac{\binom{\ell(H)}{j}}{2^j \binom{n}{j}}, \quad (2.5)$$

with $\binom{b}{j} = 0$ when $b < j$.

The comparison we need is the Boolean case of the pointwise cubical replacement principle in [7, Lemma 6.1]. It follows directly from (2.5). Put $b = \ell(H)$ and write $x_+ = \max\{x, 0\}$. For every $n \geq k$,

$$\rho_{k-1}(H)^{1/(k-1)} \leq \frac{b}{2n}, \quad \rho_k(H)^{1/k} \geq \frac{(b - k + 1)_+}{2n}.$$

Indeed, each factor $(b-i)/(n-i)$ is at most b/n and, when $b \geq k$, at least $(b-k+1)/n$. Consequently,

$$\rho_{k-1}(H)^{1/(k-1)} \leq \rho_k(H)^{1/k} + \frac{k-1}{2n}.$$

The inequality $(x + y)^k \leq 2^{k-1}(x^k + y^k)$ now gives

$$\rho_k(H) \geq 2^{1-k} \rho_{k-1}(H)^{k/(k-1)} - \left(\frac{k-1}{2n} \right)^k. \quad (2.6)$$

The additive error is necessary: if $b = k - 1$, then $\rho_k(H) = 0$ but $\rho_{k-1}(H) > 0$.

We next express the effect of shorter clauses in terms of the remaining expected time. This is the form of replacement used in the rest of the proof.

Lemma 2.1. *Fix $L < \infty$. For a satisfiable formula G on $n \geq k$ variables, let $q_G(z)$ be the probability that adjoining an independent Poisson number of mean z of uniform signed $(k-1)$ -clauses makes G unsatisfiable. There is a constant $C_{k,L}$ such that*

$$q_G(z) \leq C_{k,L} r_n(G)^{-(k-1)/k}, \quad 0 \leq z \leq L. \quad (2.7)$$

Proof. Let $H_{t,z}$ be obtained from G by adjoining independent Poisson numbers of uniform k -clauses and uniform $(k-1)$ -clauses, with means t and z , respectively. Set

$$Q(t, z) = \mathbb{P}(H_{t,z} \text{ is unsatisfiable}), \quad S(t, z) = 1 - Q(t, z).$$

Differentiating the Poisson probabilities gives

$$\begin{aligned} Q_t(t, z) &= \mathbb{E}[\mathbf{1}_{\{H_{t,z} \text{ is satisfiable}\}} \rho_k(H_{t,z})], \\ Q_z(t, z) &= \mathbb{E}[\mathbf{1}_{\{H_{t,z} \text{ is satisfiable}\}} \rho_{k-1}(H_{t,z})]. \end{aligned}$$

Here and below, a hazard multiplied by the indicator of satisfiability is defined to be zero at an unsatisfiable formula. Apply (2.6) and then Jensen's inequality to the random variable $\mathbf{1}_{\{H_{t,z} \text{ is satisfiable}\}} \rho_{k-1}(H_{t,z})$. This yields

$$Q_t \geq c_k Q_z^{k/(k-1)} - C_k n^{-k}. \quad (2.8)$$

For $v, a \geq 0$, minimizing over v gives $c_k v^{k/(k-1)} - av \geq -A_k a^k$ for a constant A_k . Fix $t > 0$ and integrate along the segment $(u, z(1 - u/t))$, $0 \leq u \leq t$. Its directional derivative is $Q_t - (z/t)Q_z$, so, since $Q(0, z) = q_G(z)$,

$$q_G(z) \leq Q(t, 0) + A_k \frac{z^k}{t^{k-1}} + C_k \frac{t}{n^k}. \quad (2.9)$$

To bound $Q(t, 0)$, put $r = r_n(G)$. The survival function is submultiplicative: $S(u + v, 0) \leq S(u, 0)S(v, 0)$. Indeed, satisfiability after adjoining both independent blocks of clauses implies satisfiability after adjoining either block to G separately. The continuous-time remaining lifetime has mean r , by conditioning on the exponential waiting times as in (2.1). Since $S(\cdot, 0)$ is nonincreasing,

$$r = \int_0^\infty S(u, 0) du \leq t \sum_{j \geq 0} S(t, 0)^j = \frac{t}{Q(t, 0)}.$$

The denominator is positive for $t > 0$, since the Poisson process has positive probability of containing every possible k -clause by time t . Thus $Q(t, 0) \leq t/r$, and (2.9) becomes

$$q_G(z) \leq \frac{t}{r} + A_k \frac{z^k}{t^{k-1}} + C_k \frac{t}{n^k}. \quad (2.10)$$

Taking $t = r^{1/k}$ balances the first two terms. For $0 \leq z \leq L$,

$$q_G(z) \leq r^{-(k-1)/k} \left(1 + A_k L^k + C_k \frac{r}{n^k} \right) \leq C_{k,L} r^{-(k-1)/k},$$

where the last inequality uses $r = O_k(n)$ from (2.3). This proves the lemma. $\square$

Remark 2.2. The shorter clauses can create new forced variables and thereby make later clauses more effective. For example, the four signed 2-clauses on two variables are jointly inconsistent, although none alone destroys the empty formula. A union bound based only on the initial value of $\rho_{k-1}(G)$ therefore cannot replace the interpolation. The differential inequality (2.8) is applied to the evolving formula, which is why it accounts for cooperation between the shorter clauses.

3. OCCUPATION ESTIMATES AND DELETION OF A VARIABLE

The remaining expected time need not decrease by a comparable amount on every step. Its conditional expected decrease is nevertheless exactly one. A concave potential converts this identity into a bound on the sum of inverse powers along the process. An expression multiplied by the indicator of satisfiability is defined to be zero at an unsatisfiable formula, including when it contains a negative power of r_n .

Lemma 3.1. *For every $\lambda > 0$,*

$$\sum_{m \geq 0} \mathbb{E} \left[\mathbf{1}_{\{T_n > m\}} r_n(F_{n,m})^{-\lambda} \right] \leq 2^\lambda \int_0^{a_n} (1+x)^{-\lambda} dx. \quad (3.1)$$

The right-hand side is $O_\lambda(a_n^{1-\lambda})$ for $0 < \lambda < 1$, $O(\log(1+a_n))$ for $\lambda = 1$, and $O_\lambda(1)$ for $\lambda > 1$.

Proof. Set $\Phi(x) = \int_0^x (1+y)^{-\lambda} dy$. This function is increasing and concave, and $\Phi(0) = 0$. At a satisfiable state G , write $r = r_n(G) \geq 1$. By (2.4) and Jensen's inequality,

$$\begin{aligned} \Phi(r) - \mathbb{E}[\Phi(r_n(G \wedge C))] &\geq \Phi(r) - \Phi(r-1) \\ &\geq (1+r)^{-\lambda} \geq 2^{-\lambda} r^{-\lambda}. \end{aligned}$$

At an unsatisfiable state both potentials are zero. Applying this estimate conditionally at each step and summing from $m = 0$ to $m = M-1$ gives

$$2^{-\lambda} \sum_{m=0}^{M-1} \mathbb{E} \left[\mathbf{1}_{\{T_n > m\}} r_n(F_{n,m})^{-\lambda} \right] \leq \Phi(a_n) - \mathbb{E}[\Phi(r_n(F_{n,M}))] \leq \Phi(a_n).$$

Let $M \rightarrow \infty$ and use monotone convergence. The stated orders follow by evaluating the integral. $\square$

Passing to Poisson time preserves these occupation sums exactly. For every nonnegative function f of the formula, independence of Π_t and the clauses gives

$$\int_0^\infty \mathbb{E}[f(F_n(t))] dt = \sum_{m \geq 0} \mathbb{E}[f(F_{n,m})], \quad (3.2)$$

because $\int_0^\infty e^{-t} t^m / m! dt = 1$. Tonelli's theorem justifies the interchange even when either side is infinite.

Write $\beta_n(t) = \mathbb{E}[\mathbf{1}_{\{F_n(t) \text{ is satisfiable}\}} \ell(F_n(t)) / n]$ for the expected proportion of forced variables, counted only before unsatisfiability. We estimate its integral by deleting one variable. Assume $n > k$, fix a variable x , and partition the clauses present at time t into those avoiding x , those containing x , and those containing $\neg x$. Poisson splitting makes these three families independent. The formula G formed by the first family has the law of $F_{n-1}((1-k/n)t)$. Each of the other two families has a Poisson number of clauses with mean $z = kt/(2n)$. After removing the literal involving x , its clauses are independent uniform signed $(k-1)$ -clauses on the other variables. Setting $x = 0$ retains the shortened positive family and satisfies the negative family automatically; setting $x = 1$ does the reverse. Conditional on a satisfiable G , the two resulting formulas are therefore independent, and each is unsatisfiable with probability $q_G(z)$.

Exactly one branch being satisfiable is equivalent to the full formula being satisfiable and forcing x . Both branches being unsatisfiable is equivalent to the full formula being unsatisfiable. By exchangeability of the variables,

$$\beta_n(t) = 2\mathbb{E} \left[\mathbf{1}_{\{G \text{ is satisfiable}\}} q_G(z)(1 - q_G(z)) \right], \quad (3.3)$$

$$s_n(t) = \mathbb{E} \left[\mathbf{1}_{\{G \text{ is satisfiable}\}} (1 - q_G(z)^2) \right]. \quad (3.4)$$

These identities use only the Poisson splitting and the two possible values of a Boolean variable.

Proposition 3.2. *For fixed $k \geq 3$,*

$$\int_0^\infty \beta_n(t) dt = O_k(n^{1/k}). \quad (3.5)$$

Proof. It suffices to consider $n \geq 2k$, since (2.3) bounds the integral by a_n for each of the finitely many remaining values. Choose a sufficiently large constant $L = L(k)$. For $0 \leq t \leq Ln$, the parameter $z = kt/(2n)$ is at most $kL/2$. Apply Lemma 2.1 on $n-1$ variables in (3.3). This gives

$$\beta_n(t) \leq C_k \mathbb{E} \left[\mathbf{1}_{\{G \text{ is satisfiable}\}} r_{n-1}(G)^{-(k-1)/k} \right].$$

Changing variables to $u = (1 - k/n)t$, then using (3.2) and Lemma 3.1, we obtain

$$\begin{aligned} \int_0^{Ln} \beta_n(t) dt &\leq \frac{C_k}{1 - k/n} \int_0^\infty \mathbb{E} \left[\mathbf{1}_{\{F_{n-1}(u) \text{ is satisfiable}\}} r_{n-1}(F_{n-1}(u))^{-(k-1)/k} \right] du \\ &= \frac{C_k}{1 - k/n} \sum_{m \geq 0} \mathbb{E} \left[\mathbf{1}_{\{T_{n-1} > m\}} r_{n-1}(F_{n-1,m})^{-(k-1)/k} \right] \\ &\leq C_k a_{n-1}^{1/k} = O_k(n^{1/k}). \end{aligned}$$

Here $1 - k/n \geq 1/2$. For $t > Ln$, use $\beta_n(t) \leq s_n(t)$ and (2.2) to get

$$\int_{Ln}^\infty \beta_n(t) dt \leq 2^k \exp \left(n \log 2 - \frac{Ln}{2^k} \right).$$

This is exponentially small when $L > 2^k \log 2$, completing the proof. $\square$

4. VARIANCE, SCALING WINDOWS, AND THE LIMITING DENSITY

We now use the occupation estimate to bound the remaining expected time at the last satisfiable formula. The link to the variance is the following general covering theorem.

Theorem 4.1 (Aldous [5], Theorem 1(b)). *Let X be a nonempty finite set, and let $Y_1, Y_2, \dots$ be independent copies of a random subset of X such that $\mathbb{P}(x \in Y_1) > 0$ for every $x \in X$. Let T be the first time these sets cover X , and let $\mathcal{T} = X \setminus \bigcup_{i < T} Y_i$ be the last uncovered set. For deterministic $A \subseteq X$, let $c(A)$ be the expected time to cover A using a fresh independent sequence with the same law. There is an absolute constant κ such that*

$$\text{Var}(T) \leq \kappa \mathbb{E} T \mathbb{E}[c(\mathcal{T})].$$

Proof of Theorem 1.3. Take $X = \{0, 1\}^n$ and let Y_i be the assignments falsifying C_i . Each assignment belongs to Y_i with probability 2^{-k} , and the resulting cover time is T_n . If $\Gamma_n = F_{n, T_n-1}$, then the terminal uncovered set is the solution set of Γ_n . Thus Theorem 4.1 gives

$$\text{Var}(T_n) \leq \kappa a_n \mathbb{E} r_n(\Gamma_n). \quad (4.1)$$

For any satisfiable G with $\rho_k(G) > 0$, waiting for a clause that by itself excludes every original solution of G shows that $r_n(G) \leq 1/\rho_k(G)$. Together with $r_n(G) \leq a_n$, this yields

$$\rho_k(G) r_n(G) \leq \min\{1, a_n \rho_k(G)\} \leq a_n^{1/k} \rho_k(G)^{1/k} \leq \frac{a_n^{1/k}}{2} \frac{\ell(G)}{n}. \quad (4.2)$$

The last inequality follows from $\binom{\ell(G)}{k} / \binom{n}{k} \leq (\ell(G)/n)^k$ when $\ell(G) \geq k$; if $\ell(G) < k$, the left-hand side is zero. The estimate therefore holds in all cases.

Decompose according to the final transition. Since $\{T_n = m+1\}$ occurs when $F_{n,m}$ is satisfiable and the next clause makes it unsatisfiable, conditioning on the first m clauses gives

$$\begin{aligned} \mathbb{E} r_n(\Gamma_n) &= \sum_{m \geq 0} \mathbb{E} \left[\mathbf{1}_{\{T_n = m+1\}} r_n(F_{n,m}) \right] \\ &= \sum_{m \geq 0} \mathbb{E} \left[\mathbf{1}_{\{T_n > m\}} \rho_k(F_{n,m}) r_n(F_{n,m}) \right]. \end{aligned}$$

Consequently, equation (4.2), the Poisson occupation identity, and Proposition 3.2 imply

$$\mathbb{E} r_n(\Gamma_n) \leq \frac{a_n^{1/k}}{2} \sum_{m \geq 0} \mathbb{E} \left[\mathbf{1}_{\{T_n > m\}} \frac{\ell(F_{n,m})}{n} \right] = \frac{a_n^{1/k}}{2} \int_0^\infty \beta_n(t) dt = O_k(n^{2/k}).$$

Using $a_n = O_k(n)$ in (4.1) proves $\text{Var}(T_n) = O_k(n^{1+2/k})$. The stated tail estimate follows from Chebyshev's inequality. $\square$

Proof of Theorem 1.1. The identity $\mathbb{P}(F_{n,m} \text{ is unsatisfiable}) = \mathbb{P}(T_n \leq m)$ identifies $\tilde{m}_u(n)$ with the corresponding quantile of T_n . Set $d = \sqrt{2 \text{Var}(T_n)/\eta}$. Chebyshev's inequality gives $\mathbb{P}(|T_n - a_n| \geq d) \leq \eta/2$, so both quantiles defining $\widetilde{W}_\eta(n)$ lie between $a_n - d$ and $a_n + d$, up to integer rounding. Thus $\widetilde{W}_\eta(n) \leq 2d + 2 = O_{k,\eta}(n^{1/2+1/k})$. $\square$

To apply the criterion for a limiting density, we transfer the estimate to μ_p on $2^{\mathcal{C}_n}$. Recall that $W_\eta(n) = N_n(p_{1-\eta}(n) - p_\eta(n))$. As in [7], let $\widehat{W}_\eta(n)$ denote the corresponding window for a uniformly chosen set of distinct clauses.

Proposition 4.2. *For every fixed $k \geq 3$ and $\eta \in (0, 1/2)$,*

$$W_\eta(n) = O_{k,\eta}(n^{1/2+1/k}). \quad (4.3)$$

The same bound holds for $\widehat{W}_\eta(n)$.

Proof. By (2.1) and Theorem 1.3, $\text{Var}(\tau_n) = O_k(n^{1+2/k})$. Its fixed-level quantile window is therefore $O_{k,\eta}(n^{1/2+1/k})$. Poisson splitting shows that at time t the multiplicities of the N_n possible clauses are independent Poisson variables of mean t/N_n . Since repetitions do not affect satisfiability, $\mathbb{P}(F_n(t) \text{ is unsatisfiable}) = \mu_{p(t)}(\text{Cover}_n)$ for $p(t) = 1 - e^{-t/N_n}$. As $N_n p'(t) = e^{-t/N_n} \leq 1$, the change of parameter does not increase the quantile window when it is measured in expected clause count. This proves (4.3).

For the model without replacement, fix a sufficiently large constant $K = K(k)$. Among the first $\lceil Kn \rceil$ independent draws, the probability of a repeated clause is at most $\binom{Kn}{2}/N_n = O_k(n^{2-k}) = o(1)$. Conditioned on no repetitions, the ordered sequence is uniform among sequences of distinct clauses. Thus the satisfiability probabilities in the two models differ by $o(1)$ uniformly for $m \leq Kn$. The discrete bound in (2.2) places all fixed central quantiles below Kn . For large n , the without-replacement window from η to $1 - \eta$ is contained between the with-replacement quantiles at levels $\eta/2$ and $1 - \eta/2$. Theorem 1.1 now gives the assertion. $\square$

We use the Abbe–Montanari criterion [1] in the form stated in [13, Proposition 14]: if, for some fixed $\delta > 0$ and every fixed $\eta \in (0, 1/2)$, the independent-inclusion window satisfies $W_\eta(n) = O_{k,\eta}(N_n p_{1/2}(n)/(\log n)^{1+\delta})$, then the satisfiability threshold has a limiting clause density. The standard linear bounds on the critical number of clauses give $N_n p_{1/2}(n) = \Theta_k(n)$; see, for example, [7, Corollary 7.6] with alphabet size two. Thus the criterion has exactly the expected-clause-count normalization used in (1.1).

Proof of Corollary 1.2. Since $1/2 + 1/k < 1$, Proposition 4.2 gives $W_\eta(n) = O_{k,\eta}(n^{1/2+1/k}) = o(n/(\log n)^2)$. The Abbe–Montanari criterion, with $\delta = 1$, therefore yields a limiting density $\alpha_k \in (0, \infty)$ in the independent-inclusion model.

For clarity, we give the transfer back to a fixed number of independent clauses. At time $t = \alpha n$, the corresponding independent-inclusion model has expected clause count $N_n p(\alpha n) = \alpha n + O_k(n^2/N_n) = \alpha n + o(1)$. Monotonicity therefore gives $s_n(\alpha n) \rightarrow 1$ for fixed $\alpha < \alpha_k$, and $s_n(\alpha n) \rightarrow 0$ for fixed $\alpha > \alpha_k$. If $0 \leq \alpha < \alpha_k$, choose $\alpha' \in (\alpha, \alpha_k)$ and couple $F_{n, \lfloor \alpha n \rfloor}$ and $F_n(\alpha' n)$ using the same clause sequence. Then

$$\mathbb{P}(F_{n, \lfloor \alpha n \rfloor} \text{ is unsatisfiable}) \leq \mathbb{P}(F_n(\alpha' n) \text{ is unsatisfiable}) + \mathbb{P}(\Pi_{\alpha' n} < \lfloor \alpha n \rfloor) \rightarrow 0.$$

If $\alpha > \alpha_k$, choose $\alpha' \in (\alpha_k, \alpha)$ and similarly obtain

$$\mathbb{P}(F_{n, \lfloor \alpha n \rfloor} \text{ is satisfiable}) \leq s_n(\alpha' n) + \mathbb{P}(\Pi_{\alpha' n} > \lfloor \alpha n \rfloor) \rightarrow 0.$$

The Poisson error probabilities vanish because their means and the comparison counts differ by order n . This proves the corollary. $\square$

The mean hitting time converges to the same limiting density, and the variance estimate gives a polynomial error probability at any fixed positive distance from it.

Corollary 4.3. *We have $a_n/n \rightarrow \alpha_k$. Moreover, for every fixed $d \in (0, \alpha_k)$,*

$$\begin{aligned} & \mathbb{P}(F_{n, \lfloor (\alpha_k - d)n \rfloor} \text{ is unsatisfiable}) \\ & + \mathbb{P}(F_{n, \lceil (\alpha_k + d)n \rceil} \text{ is satisfiable}) = O_{k,d}(n^{-1+2/k}). \end{aligned}$$

Proof. Corollary 1.2 gives $T_n/n \rightarrow \alpha_k$ in probability. Theorem 1.3 gives $(T_n - a_n)/n \rightarrow 0$ in probability, so the deterministic sequence a_n/n also converges to α_k . For all sufficiently large n , $|a_n/n - \alpha_k| < d/2$. Each event in the displayed estimate then implies a deviation of T_n from a_n by at least $dn/2$. Applying Theorem 1.3 completes the proof. $\square$

5. REMARKS AND QUESTIONS

5.1. The lower bound and the possible window exponent. Wilson [14, Corollary 4] proved $\widetilde{W}_\eta(n) = \Omega_{k,\eta}(n^{1/2})$ for every fixed k and $\eta \in (0, 1/2)$. This holds when the number of clauses is fixed, both with and without replacement. It is therefore not merely a consequence of fluctuations in the clause count of a Poisson or independent-inclusion model. The proof uses clauses containing a variable that occurs nowhere else in the formula: such clauses can be removed without changing satisfiability.

Together with Theorem 1.1, this gives

$$c_{k,\eta} n^{1/2} \leq \widetilde{W}_\eta(n) \leq C_{k,\eta} n^{1/2+1/k}.$$

For 3-SAT, the bounds are $\Omega_\eta(n^{1/2})$ and $O_\eta(n^{5/6})$. The present proof does not establish that the upper exponent is sharp. It records the cost of two estimates, neither of which is shown to be attained along the random process.

Question 5.1. For every fixed $k \geq 3$ and $\eta \in (0, 1/2)$, is $\widetilde{W}_\eta(n) = O_{k,\eta}(n^{1/2})$? More modestly, is $\widetilde{W}_\eta(n) = n^{1/2+o(1)}$?

The lower bound alone gives no reason that equality should hold. In particular, the 2-SAT window has the larger order $\Theta(n^{2/3})$ [6]. A square-root upper bound for $k \geq 3$ would need additional information about the process, not just its monotonicity.

5.2. Where the two factors of $n^{1/k}$ enter. Two bounds contribute to the terminal expectation. The occupation estimate gives $\int_0^\infty \beta_n(t) dt = O_k(n^{1/k})$, and the passage to terminal sampling gives

$$\mathbb{E}r_n(\Gamma_n) \leq \frac{a_n^{1/k}}{2} \int_0^\infty \beta_n(t) dt.$$

Thus $\mathbb{E}r_n(\Gamma_n) = O_k(n^{2/k})$ has two separate sources.

The first is the occupation bound. Replacement gives the inverse power $r^{-(k-1)/k}$, and the potential argument sums it at a cost $\int_0^{a_n-1} (1+u)^{-(k-1)/k} du = O_k(n^{1/k})$. The bound uses the initial remaining lifetime and its conditional drift, but not a more detailed description of the states visited by the process. A smaller accumulated contribution could therefore hold for random formulas even if the replacement estimate is optimal for arbitrary residual solution sets.

The second factor arises in passing to the terminal distribution. We use only

$$\rho_k(G)r_n(G) \leq \min\{1, a_n\rho_k(G)\} \leq a_n^{1/k}\rho_k(G)^{1/k} \leq \frac{a_n^{1/k}}{2} \frac{\ell(G)}{n}.$$

The scalar inequality $\min\{1, x\} \leq x^{1/k}$ is sharp at $x = 1$. This does not show that states near equality carry a substantial part of the occupation measure. In particular, the estimate discards information about the joint behaviour of $\rho_k(G)$ and $r_n(G)$, and about the further forcing created by subsequent clauses. The upper bound on $\mathbb{E}r_n(\Gamma_n)$ should not be read as a determination of a typical terminal lifetime, or as a count of the last surviving assignments.

There is also a limit to sharpening the pointwise comparison alone. Consider a subcube with b fixed coordinates, where $b \rightarrow \infty$ and $b/n \rightarrow 0$. For its solution set, equation (2.5) gives $\rho_j \asymp_k (b/n)^j$ for $j \in \{k-1, k\}$. Thus the power $k/(k-1)$ in the comparison of ρ_k with ρ_{k-1} cannot be decreased uniformly over arbitrary residual solution sets. This does not establish sharpness of the replacement lemma or of the window bound. It suggests looking for information about which states the clause process visits, or how it moves between them, rather than only improving the one-step comparison.

5.3. A question about the last surviving solutions. The covering inequality provides a concrete sufficient target for a square-root window.

Question 5.2. For each fixed $k \geq 3$, is $\sup_{n \geq k} \mathbb{E} r_n(\Gamma_n) < \infty$? More modestly, does $\mathbb{E} r_n(\Gamma_n) = n^{o(1)}$ hold?

A bounded terminal expectation would give $\text{Var}(T_n) = O_k(n)$, and hence an upper bound matching Wilson's lower bound. A subpolynomial terminal expectation would give $\widetilde{W}_\eta(n) = n^{1/2+o(1)}$. These are sufficient conditions; the corresponding fixed-level window estimates need not by themselves control all the tails required for a variance bound.

The distinction between an expectation and a high-probability statement is important here. For example, suppose that $\ell(\Gamma_n) \geq cn$ outside an event of probability δ_n , for some fixed $c > 0$. On that event's complement, the one-clause destruction probability is bounded away from zero, so $r_n(\Gamma_n) = O_{k,c}(1)$. On the exceptional event we have only $r_n(\Gamma_n) = O_k(n)$. These two bounds give $\mathbb{E} r_n(\Gamma_n) = O_{k,c}(1) + O_k(n\delta_n)$. Thus a qualitative high-probability assertion would not alone prove a bounded terminal expectation. No such assertion about $\ell(\Gamma_n)$ is needed in the present argument.

The exact terminal identity is

$$\mathbb{E} r_n(\Gamma_n) = \sum_{m \geq 0} \mathbb{E} \left[\mathbf{1}_{\{T_n > m\}} \rho_k(F_{n,m}) r_n(F_{n,m}) \right].$$

It retains the weighting by the probability of destruction, which is lost when we bound the summand by a multiple of $\ell(F_{n,m})/n$. Estimating this weighted sum directly is one possible route to a sharper result.

5.4. The change in the mean when a variable is added. The second deletion identity also bounds the change in the mean hitting time when one variable is added.

Corollary 5.3. For fixed $k \geq 3$ and $n > k$,

$$0 \leq \frac{a_{n-1}}{1 - k/n} - a_n \leq C_k. \quad (5.1)$$

In particular, $|a_n - a_{n-1}| = O_k(1)$.

Proof. By (3.4),

$$s_{n-1}((1 - k/n)t) - s_n(t) = \mathbb{E} \left[\mathbf{1}_{\{G \text{ is satisfiable}\}} q_G(kt/(2n))^2 \right] \geq 0.$$

Its integral is precisely the difference in (5.1), since $\int_0^\infty s_n(t) dt = a_n$. Choose a sufficiently large constant $L = L(k)$. For $n \geq 2k$ and $0 \leq t \leq Ln$, Lemma 2.1 bounds the integrand by $C_k \mathbb{E} [\mathbf{1}_{\{G \text{ is satisfiable}\}} r_{n-1}(G)^{-2(k-1)/k}]$. Since $2(k-1)/k > 1$, the change of variables and occupation estimate used in Proposition 3.2 give a bound independent of n . For $t > Ln$, the difference is at most $s_{n-1}((1 - k/n)t) \leq 2^{n-1} e^{-(1-k/n)t/2^k}$, whose integral is exponentially small for sufficiently large L . Increasing the constant handles $k < n < 2k$. Finally,

$$a_n - a_{n-1} = \frac{ka_{n-1}}{n - k} - \left(\frac{a_{n-1}}{1 - k/n} - a_n \right) = O_k(1)$$

by (2.3) and the bound just proved. $\square$

Statement regarding artificial intelligence. None of the mathematical arguments in this paper was generated by a large language model.

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DEPARTMENT OF PURE MATHEMATICS AND MATHEMATICAL STATISTICS, UNIVERSITY OF CAMBRIDGE, CAMBRIDGE, UNITED KINGDOM

Email address: `gc645@cam.ac.uk`